

SINGH NITIN RAKESH

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Professional Summary

AI Engineer with 1.8+ years of experience in developing LLM-powered applications, RAG pipelines, and intelligent workflows using **Python**, **FastAPI**, **LangChain**. Specialized in Agentic AI workflows. Skilled in fine-tuning models (LoRA), optimizing inference pipelines, building multi-agent systems for automated decision-making, and deploying scalable solutions on **AWS**.

Education

Uka Tarsadia University | Surat, Gujarat

B.Tech Artificial Intelligence & Data Science

Agg. CGPA: 8.68

Sep 2021 - May 2025

Achieved a perfect 10.0 SGPA in the final semester, demonstrating mastery of advanced coursework.

Skills

Deep Learning & CV : PyTorch, TensorFlow, Stable Diffusion, LoRA Fine-Tuning, Automatic1111, Kohya SS, Computer Vision

Generative AI & LLMs : OpenAI API, Google Gemini, LangChain, RAG Pipelines, MCP, Agentic AI, ADK, SDK, Multi-Agent Systems

Languages & Databases : Python (Advanced), React.js, PostgreSQL, Vector Databases (ChromaDB/Pinecone/FAISS), MongoDB

Cloud & Tools : Amazon Web Services (AWS), Docker, Git, Streamlit, Flask, FastAPI

Soft Skills : Communication, Discipline, Problem Solving, Leadership, Consistency, Proactiveness, Reliability,

Projects

Multi-Agent Competitive Querying System (AMD-AI-League) | Agentic AI & LLM Orchestration

[GitHub Link](#)

- Architected a Multi-Agent System comprising distinct "Questioner" and "Answerer" agents to simulate competitive information retrieval scenarios.
- Implemented complex prompting strategies where the Question Agent generates context-aware queries and the Answer Agent retrieves and synthesizes data to respond.
- Designed the evaluation logic to score agent performance based on accuracy and relevance, laying the groundwork for automated negotiation systems.

RAG-Based Financial Intelligence Agent | Data & Gen AI, RAG & Flask

[GitHub Link](#) | [Live Link](#)

- Developed a Retrieval-Augmented Generation (RAG) pipeline that aggregates real-time financial news from Polygon and Finviz APIs.
- Optimized the RAG pipeline using semantic chunking strategies and vector embeddings to manage context window limits, ensuring high-accuracy retrieval for Gemini Pro.
- Deployed the solution using Flask, handling dynamic context injection to ensure the LLM responses were grounded in live market data, eliminating hallucinations.

Generative AI Image Pipeline & LoRA Fine-Tuning | Deep Learning & Computer Vision

[GitHub Link](#)

- Engineered an end-to-end image generation pipeline (AI-Avtaar) utilizing Stable Diffusion and Automatic1111 for high-fidelity character creation.
- Fine-tuned custom LoRA models and applied 4-bit Quantization techniques to optimize model inference latency and reduce memory usage during deployment.
- Integrated a Virtual Try-On module using Computer Vision (CatVTON), allowing users to overlay generated clothing onto custom avatars seamlessly.

Solana Market Analysis Pipeline | Data Engineering & Analytics

[GitHub Link](#) | [Live Link](#)

- Managed an end-to-end ETL pipeline processing 1,300+ days of crypto market data.
- Enriched datasets with 25+ technical indicators (RSI, MACD) to feed into predictive models, demonstrating strong data engineering capabilities required for AI pipelines.

Experience

AI/ML Engineer | ELaunch Solution Pvt. Ltd., Surat, Gujarat

Dec 2024 - Present

- Developing an **AI-driven game-personalization-engine** utilizing Behavioral Session Telemetry and reinforcement learning (RTP, game volatility) to maximize player engagement in real-time.
- **Fine-tuned a Temporal Fusion Transformer** model for crypto price forecasting by engineering a robust pipeline with 25+ technical indicators across 1,300+ days of Solana data.
- Built and deployed 3+ **Naive RAG** and **Agentic RAG** pipelines, and developed production-ready AI chatbots end-to-end.

Data Analyst Intern | JBcodeapp, Surat, Gujarat

May 2024 - June 2024

Project: Student Performance Analysis | Python, NumPy

- **Engineered** an automated data pipeline using **Python** to perform ETL (Extract, Transform, Load) on **5,000+ student records**.
- **Conducted** exploratory data analysis to identify the key factors and patterns influencing student academic performance.

AI/ML Trainee | AlgoBrain AI, Surat, Gujarat

Sep 2023 - Jan 2024

Project: Movie-Ticket Booking Chatbot with DialogFlow CX

- **Analyzed** potential user conversation paths to design and build a movie-ticket booking chatbot in Dialogflow CX, enhancing the overall user experience.

Extracurricular Activities

Research Community Member Swaayan Drone Club, SVNIT, Surat, Gujarat	May 2024 - June 2024
• Built and Assisted with Drone Bootcamps by building and calibrating drones using Betaflight and Mission Planner.	
User Software Tester (Part-Time) Nothing India, Remote	Feb 2023 - Present
• Conducted beta testing for Nothing Phone 1, Nothing Phone 2, Nothing 2a, Android 14, CMF Watch, and more.	
Aspire Leaders Program Aspire Institute, Harvard Business School, Online	Aug 2024 - Nov 2024
• Completed all modules in Leadership, Communication, and Problem-Solving.	
Google Student Ambassador Program Google, Ping Network MCN, Online	Sep 2025 – Dec 2025
• Selected as the Google Student Ambassador from my college, serving as the campus representative for Google .	
Minimax Ambassador Minimax.io, Remote	Jan 2026 – Present
• Selected as Minimax Ambassador to test the model and give feedback.	

Achievements

- Awarded a **Certificate of Excellence** in Summer Analytics (IIT Guwahati) for finishing in the top 10 of all participants.
- Certified in **Time Series Analysis** by **IIT Guwahati**, with demonstrated expertise in statistical forecasting models.
- Achieved **Runner-Up** position in Tattva Hackathon 2025 (CGPIT), demonstrating problem-solving, teamwork, and innovation.
- Achieved **Winner** position in Quiz (Group) Competition at College Fest 2023, showcasing knowledge, teamwork, and quick thinking.
- Secured **2nd Runner-Up** position in Script Writing (Solo) at College Fest 2023, demonstrating creativity and communication skills.
- Secured **Top 5 finishes at IITGN (Finnovate/Amalthea) and IIT Bombay (AMD) Hackathons**, recognized for technical excellence and innovative problem-solving.

Certifications

- Oracle AI Vector Search Certified Professional [\[View Credential\]](#)
- Natural Language Processing in TensorFlow [\[View Credential\]](#)
- Retrieval Augmented Generation (RAG) [\[View Credential\]](#)
- Fine Tune BERT for Text Classification with TensorFlow [\[View Credential\]](#)
- Machine Learning Specialization [\[View Credential\]](#)
- Completed ML: An Overview from Coursera [\[View Credential\]](#)

Publications

Is Architectural Complexity Always the Answer? A Case Study on SwinIR vs. an Efficient CNN Chandresh Sutariya, Nitin Singh
• Abstract: It has been demonstrated that an optimized CNN achieves near state-of-the-art restoration quality, comparable to SwinIR, while significantly reducing computational overhead.
• arXiv Pre-print, 2025